

Shanu Dengre

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Summary

Customer-facing AI Engineer and Data Scientist with 4+ years of taking ML systems from problem scoping to production. Builds forecasting, computer vision and LLM-based solutions tuned to each customer's business constraints, deployed across 300+ locations. Improved forecast accuracy by 20% and cut food waste by 15%. PhD in Physics.

Skills

AI & Deep Learning: PyTorch, TensorFlow, Scikit-learn, Transformers

GenAI / LLMs: LLM APIs, Agentic Workflows, Prompt Engineering, RAG

Time Series Forecasting: DARTS, Nixtla, CHRONOS (Foundation Model)

Computer Vision: YOLO-family, OpenCV, SAM

Software & Deployment: Python, FastAPI, Docker, Kubernetes, Git/GitLab, CI/CD

Data Engineering & MLOps: PySpark, MongoDB, Snowflake, SQL, AWS, MLflow, Sentry

Analytics & Delivery: Pandas, NumPy, PowerBI, Scrum, Jira, Confluence

Languages: English (Fluent), German (A2.2), Hindi (Native)

Experience

InterProductTec – Mumbai

Nov 2025 – Present

Sr. Data Scientist

Demand Forecast Project Lead

- Lead the demand forecast project end to end, from customer scoping to features in production.
- Deployed automated POS integration pipelines for faster customer onboarding to forecast solutions.
- Developed an in-house newsvendor-based ordering optimisation balancing margin against stockout risk, tuning forecast quantiles to each customer's requirements to minimise food waste without losing sales.
- Innovated forecasting solutions improving sales forecast by 20% during holiday/offer seasons.
- Enhanced pipeline observability with Sentry and PowerBI dashboards, achieving negligible downtime while delivering SOTA forecast accuracy across 300+ locations.
- Built LLM-powered agentic workflows (tool-calling agents) to automate internal analytics and reporting.
- Automated customer food waste reports, speeding up delivery and reducing manual effort by 50%.

PreciTaste – Munich

Feb 2022 – Oct 2025

AI Engineer and Data Scientist

Time Series Forecasting – Sales and Demand

- Led AI-driven demand forecasting, reducing food waste by 15% and improving crew efficiency by 20%.
- Designed an automated forecasting pipeline using DARTS, cutting retraining time by 50%.
- Reduced query latency by 60% using FastAPI microservices with MongoDB, Docker, and Kubernetes.

Computer Vision – Object Detection and Segmentation

- Deployed AI-based food recognition using YOLO-family and PyTorch across 10+ restaurant chains.
- Built automated annotation pipelines with OpenCV and SAM, reducing labeling time by 50%.
- Delivered a POS order-matching system with 95%+ real-time validation accuracy.

Apps

Newsvendor Problem Visualiser — [🔗 newsvendorproblem.streamlit.app](#)

Python, Streamlit

Interactive newsvendor simulator: weigh stockout vs. overstock losses and find the best safety margin.

Education

Data Science & Machine Learning Bootcamp, Le Wagon Berlin

June 2021 – Sept 2021

PhD (Physics), TU Dresden — Publication: Phys. Rev. B 103, 064425 (2021)

Oct 2015 – Feb 2021

BSc–MSc (Physics), IISER Pune — Publication: Phys. Rev. B 95, 054401 (2017) Aug 2010 – May 2015